

Joy D'Souza

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EDUCATION

Georgia Institute of Technology (GT), Atlanta, GA

August 2024 – May 2028

Bachelor of Science in Mechanical Engineering; GPA: 4.0/4.0

Activities: Engineers Without Borders (Recruitment Co-lead), Medical Robotics, Society of Women Engineers (SWE), Chinese Student Association, American Society of Mechanical Engineers

WORK EXPERIENCE

Design for Planet Impact, *Student Assistant*, Atlanta, GA

June 2026 – Present

- Researched global challenges aligned with UN Sustainable Development Goals for a student-led organization developing sustainable, human-centered devices for specific communities; investigated existing solutions, adoption drivers and barriers, and target markets and audiences to identify opportunities and propose device concepts for development and prototyping.

SharkNinja, *Product Test Engineering Intern*, Needham, MA

May 2026 – July 2026

- Utilized Claude to develop a Python-based annotation application and standardized measurement methodology to objectively quantify hair curl geometry through six geometric metrics and correlated measurements with consumer ratings to establish KPI target zones for product performance. Authored tool work instructions, trained technicians on the measurement protocol, and conducted Gage R&R analysis to determine inter-operator repeatability and assess measurement-system suitability for future testing.
- Designed and assembled a standardized curl-retention test rig for 15 hair tresses, establishing camera alignment, lighting, and styling procedures for repeatable testing; used Claude to develop a web application that generated time-lapse visualizations and quantified curl length changes.
- Reviewed 50+ completed hair care test reports such as coanda force, frizz and shine, and straightness and alignment to verify key metrics fell within established ± 2 standard deviation limits and confirm data integrity before releasing results for cross-functional review; performed one-way ANOVA in Minitab to assess statistical parity of key metrics between test results.
- Created deterministic Python extraction tools using Claude to pull core requirements and data from any format of Design Qualification Test Plan (DQTP) or Design Failure Mode and Effects Analysis (DFMEA) document, incorporating an LLM API planning layer that analyzes DFMEA column headers, maps them to target fields, and generates an extraction plan for the deterministic parser to improve accuracy across varied formats.

Georgia Tech, *Undergraduate Researcher*, Atlanta, GA

January 2026 – Present

- Collaborated in a 2-person team to integrate a Raspberry Pi 5, machine vision system, and custom mechanical gripper with a UR5 robotic arm for industrial pick-and-place applications.
- Utilized RoboFlow to label and train object detection model on 70+ images for test tube location and color identification.
- Designed 3D-printed claw modification for existing gripper mechanism to improve object grasping performance.

SharkNinja, *Quality Engineering Intern*, Needham, MA

May 2025 – August 2025

- Analyzed 2000+ customer reviews and troubleshooting call data to identify top product detractors and delighters across various home environment units; tracked week-over-week sentiment trends to inform product improvement priorities.
- Performed hands-on teardowns and root cause analysis (RCA) on 50+ returned units to test and identify failures with PCBAs, motors, and other internal mechanisms; documented findings, testing methods, and corrective actions.
- Conducted global cross-functional collaboration with engineering, manufacturing, marketing, product development, and customer experience teams to contribute to corrective actions, including a rework to enhance RF signal strength between the remote and receiver on a key home environment product.

Mathnasium, *Level 2 Instructor*, Hanover, MA

August 2022 – August 2024

- Tutored up to three students simultaneously in elementary through tenth-grade math, tailoring lesson plans to diverse learning styles and tracking progress through performance monitoring and parent communication.

PROJECTS

Force Sensor for 3D-Printed Prosthetic, *Medical Robotics Club*, Atlanta, GA

September 2025 – Present

- Developing Force Sensing Resistor (FSR) calibration SOP to standardize sensor variability for force sensor integration into low-cost 3D-printed prosthetic for trans-radial amputees.
- Calibrating FSRs by applying known weights, collecting analog voltage data via Arduino, and analytically defining touch thresholds (soft, medium, hard) as prosthetic feedback for the user.
- Designed a 3D-printed force distributor to evenly concentrate applied loads onto the FSR sensing surface for repeatable calibration testing.

Sustainable Menstrual Pad, *Georgia Tech Engineers without Borders Wish For WASH*, Atlanta, GA

October 2024 – Present

- Collaborating with a 14-member bioplastics team to develop an adhesive-free, sustainable menstrual pad matching the absorbency, comfort, and cost of commercial standards (benchmarked against Always pads).
- Conducted single-material absorbance testing on 9 materials per ASTM and ISO standards using water and gelatin blood simulant, identifying Himalayan nettle and wood pulp as highest mass-gain core materials.
- Progressing to composite and rewet testing to evaluate candidate materials across pad layers for absorbency, permeability, and skin comfort.

Autonomous Competition Robot, *ME2110*, Atlanta, GA

September 2025 – November 2025

- Collaborating in 4-member team to design and fabricate a fully-autonomous robot in 8-weeks under a \$120 budget using provided mechatronics components and Arduino Uno for competition against 72 teams.
- Utilized Onshape, Inkscape, and Bambu Studio for component fabrication on Trotec Laser machines and BambuLab P1S printers.

- Created multiple technical reports and presentations detailing concept generation and evaluation, quality function deployment, and design overview utilizing engineering figures such as Morphological Chart, Engineering Specification Sheet, and House of Quality.

SKILLS

Technical: Technical Writing, Root Cause Analysis (RCA), 8D Problem Solving, Failure Mode and Effects Analysis (FMEA)

Software: SolidWorks, Onshape, Bambu Studio, Inkscape, Trotec JobControl, Jira, Zendesk, Google Suite, Microsoft Office, Minitab

Programming Languages: MATLAB, Java, Python, HTML

Machines: Laser Cutter/Engraver, Band Saw, Miter Saw, Soldering, Instron, Mark-10

Concepts: Differential Equations, Statics, Dynamics, Materials Selection, Deformable Bodies, Thermodynamics, Circuit Analysis, Numerical Methods

HONORS

- New England School Development Council Award for Academic Growth and Student Leadership in Learning
- Rochester Institute of Technology Computing Award

October 2023

June 2023